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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

EC2004-1 – MICROCONTROLLERS
(For ECE, ACT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The value in register A after the execution of the following instruction,
ANL A, #FFH
A) It will retain the Value of A
B) It will retain the previous value of higher nibble
C) Value of A is set to 00H
D) It will mask the value of lower nibble
2. If we say microcontroller is 8-bit then here 8-bit denotes size of:
A) System bus
B) ALU data bus
C) Control Bus
D) Address Bus
3. TF1, TR1, TF0, TR0 bits are of which register?
A) TMOD
B) SCON
C) TCON
D) SMOD
4. CLR 00H instruction in 8051 microcontroller will
A) make R0 register zero
B) makes the 00H RAM location zero
C) clears the first LSB of 20H memory location
D) clears the first LSB of 00H memory location
5. JBC in 8051 defines it as it jumps if the bit is set and ____ the bit?
A) Moves
B) Clears
C) Stands
D) None Of these
6. SETB instruction in 8051 uses operand as
A) 8 bit address
B) immediate value
C) register
D) 16 bit address
7. **MOV A, #09**
MOV B, #08
MUL AB
DA A
What is the result in A register after the execution of above code ?
A) 0X48
B) 0X72
C) 0X00
D) 0X09
8. What is the time taken by one machine cycle if crystal frequency of 8051 is 12MHz?
A) 1 micro sec
B) 12 micro sec
C) 1.085 micro sec
D) 10 micro sec

9. **MOV A,#47H**
RR A
RR A
MOV R0, A
 What is the result in R0 after the execution of above code ?
 A) 0XD1 B) 0x47
 C) 0xB3 D) 0xC3
10. The count value to be loaded in 8051 Timer in Mode1 to get maximum delay
 A) FFFF B) 0F
 C) 0000 D) FF
11. What is the value to be loaded into TMOD register to make Timer0 in mode2 and Timer1 in mode1?
 A) 21 B) 12
 C) 22 D) 11
12. If the total machine cycle required is 23000 then the counter value required for timer 1 in mode 1 is
 A) B628 B) A628
 C) A620 D) FFFF
13. The Unsigned Character data type is in the range
 A) 0-655350 B) -255
 C) -128 to +127 D) -32768 to +32767
14. What are the content of the registers r0 and r1 after the execution of following ARM instruction
LDR r0, [r1, #4]
 If r0 = 0x00000000
 r1 = 0x00090000
 mem32[0x00090000] = 0x01010101
 mem32[0x00090004] = 0x02020202
 A) r0 = 0x01010101 B) r0 = 0x02020202
 r1 = 0x00090000 r1 = 0x00090004
 C) r0 = 0x02020202 D) r0 = 0x01010101
 r1 = 0x00090000 r1 = 0x00090004
15. What are the content of the registers r0 and r1 after the execution of following instruction
LDR r0, [r1], #4
 If r0 = 0x00000000
 r1 = 0x00090000
 mem32[0x00090000] = 0x01010101
 mem32[0x00090004] = 0x02020202
 A) r0 = 0x02020202 B) r0 = 0x01010101
 r1 = 0x00090000 r1 = 0x00090000
 C) r0 = 0x02020202 D) r0 = 0x01010101
 r1 = 0x00090004 r1 = 0x00090004
16. Thumb instruction mode in ARM processor uses
 A) 24 bit code size B) 32 bit code size
 C) 8 bit code size D) 16 bit code size

17. What is the result after the execution of following instruction in r0 ?
 BIC r0, r1, r2
 If r0 = 0b0000
 r1 = 0b1111
 r2 = 0b0101
 A) r0 = 0b0101 B) r0 = 0b1111
 C) r0 = 0b1010 D) r0 = 0b0000
18. What is the result after the execution of following instruction in r0 ?
 MLA r0, r1, r2, r3
 If r0 = 0x00000001
 r1 = 0x00000002
 r2 = 0x00000003
 r3 = 0x00000004
 A) r0 = 0x00000010 B) r0 = 0x0000000a
 C) r0 = 0x0000000c D) r0 = 0x00000024
19. What is the result after the execution of following instruction in r7 ?
 MOV r7, r5, LSL #2
 r5 = 0x00000005
 r7 = 0x00000008
 A) r7 = 0x0000000C B) r7 = 0x00000020
 C) r7 = 0x00000014 D) r7 = 0x00000040
20. Which one of the following is valid for ARM for loading an immediate data in to a registers
 A) Most of the immediate values are loaded using barrel shifter B) instructions itself frames 32 bit immediate vale
 C) No immediate data is valid in ARM D) A separate 32 bit data is subsequently stored in the program memory along with 32 bit opcode

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) List and explain any four addressing modes of 8051 microcontroller.	8	L*2	1	1
b) Illustrate the working of followings instructions with an example i) DA A ii) ADD iii) MOVC iv) MOVX v) ROR	8	L2	1	1
2. a) Write an assembly language program for 8051 to find the largest number in a array of data stored from 0x10 to 0x19 memory location of internal RAM.	8	L2	1	1
b) What are the different processes involved in the execution of LCALL and RET instruction of 8051 microcontroller. Illustrate with an example.	8	L2	1	1
3. a) Write an assembly program for 8051 microcontroller to implement the following BCD arithmetic function. $f(x,y) = xy(x+y)$ where x and y are the single digit BCD number.	8	L2	1	1
b) What is bit addressable memory in 8051 microcontroller? How bit addressable instructions can be used to manipulate the memory mapped IO registers?	8	L2	1	1

Unit – II

4. a) What are the steps involved programming 8051 timer in MODE 1 ? Provide an example embedded C code to configure timer in MODE 1. 8 L2 2 1
- b) Interface a Seven segment LED to make up count from 0 to 9 with 1 second delay. Assume all the required parameters for the design. 8 L3 3 1
5. a) Provide details of the internal functional block of the timer used in 8051 microcontroller. 8 L2 2 1
- b) A personal computer (PC) UART port is connected to 8051 UART. PC is continuously sending the data sequence "\$hello#" in the ASCII format. Write a C code for 8051 to receive the data in to an array. 8 L3 3 1
6. a) What are the different programming steps to be used to transmit and receive data byte using serial communication protocol (UART) in 8051 microcontroller? 8 L2 2 1
- b) Interface 2 buttons to 8051 IO PORT and generate the square wave signal with different frequency as shown in the table 6a.

Table 6a

Button 1	Button 2	Signal frequency
0	0	1kHz , 50% duty cycle
0	1	2kHz, 25% duty cycle
1	0	4kHz, 75% duty cycle
1	1	6kHz, 50% duty cycle

Unit – III

7. a) Provide details of 3 – stage pipeline in ARM. Illustrate the 3 – stage pipeline using an ARM instruction. 8 L2 4 1
- b) Explain the functionality of following ARM instructions with an example
- MLA
 - BIC
 - RSB
 - MOV
8. a) What will be the content in the register R0 after the execution of each one of the following ARM instruction. 8 L2 5 1
- Assume that r1=0x18000014
r2=0x28000024
- MOV r0, r2, LSL #02
 - ADD r0,r1,r2,LSR #04
 - MVN r0, r1, LSL #04
- b) Illustrate the execution of branch instructions in ARM with an example. How R14 (link register) changes during the branch instruction? 8 L2 4 1